

## PRT661 – Assessment 2

### Housing Affordability Forecasting System for Greater Darwin Region

Group: Dan1-Theme2 — Prepared by Bipana Tripathee (S388875)

## 2.1 Changes Since Assessment 1

| Area                           | Previous State (A1)                                                  | Current State (A2)                                                                                                                                                                                                                                                                                 | Justification                                                                                                                                                                  |
|--------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Income / household data source | ABS Census 2021 (single-year snapshot, suburb level)                 | ABS Annual Household Income (NT-wide), continuous series covering 1990–2025                                                                                                                                                                                                                        | 2021 single-year snapshot mismatched with multi-year property sales; replaced with a continuous annual series so every sale year matches directly, with no gap-filling needed. |
| Cloud storage technology       | Generic "Cloud Database" (unnamed in A1 architecture diagram)        | AWS S3 for raw data storage                                                                                                                                                                                                                                                                        | Addresses A1 feedback that technology choices were generic and unjustified; AWS S3 is now explicitly named and used.                                                           |
| Modelling scope                | Regression and classification both implied without a stated priority | Primary focus on regression (property price as continuous target); affordability classification treated as a secondary/stretch component                                                                                                                                                           | Addresses A1 feedback that modelling scope was unclear; ensures consistency across objectives, architecture and workflow.                                                      |
| Model training and evaluation  | Not yet trained (A1 was planning only)                               | LR, Random Forest and XGBoost trained on a time-based split. Two data leakage issues identified and corrected: price_per_sqm excluded as a feature (derived directly from the target price); imputation now performed after the train/test split. Current baseline: $R^2$ 0.50–0.56 across models. | Corrects two issues that were inflating earlier results; current figures reflect genuine forecasting performance rather than in-sample accuracy.                               |
| Population data source         | ABS Census population data, grouped by SA2 region                    | ABS Estimated Resident Population (ERP), Greater Darwin region-wide                                                                                                                                                                                                                                | SA2 region names and suburb names could not be reliably matched; ERP is region-wide and merges cleanly on year with no suburb-matching required.                               |
| Income data coverage           | Not addressed in Assessment 1                                        | Household income series covers 1990–2025; any sale year without a match is automatically flagged in an audit trail rather than silently dropped or assumed                                                                                                                                         | Ensures transparency — coverage gaps are explicitly reported and checked, not hidden.                                                                                          |